

DIVERGENCE

If $\vec{F}(x, y, z) = f(x, y, z)\hat{i} + g(x, y, z)\hat{j} + h(x, y, z)\hat{k}$
then we define the divergence of $\vec{F}$
written as $\text{div } \vec{F}$ to be the function
given by

$$\text{div } \vec{F} = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z}$$

Note :- $\text{div } \vec{F}$ can also be represented
as $\boxed{\text{div } \vec{F} = \nabla \cdot \vec{F}}$

Note :- Divergence relates the way in
which fluid flows towards or away from
a point.

Note :- If $\text{div } \vec{F} = \nabla \cdot \vec{F} = 0$, then we say
that $\vec{F}$ is solenoidal.

Q Find $\text{div } \vec{F}$ where $\vec{F} = e^{xy}\hat{i} - 2\cos y\hat{j} + \sin^2 z\hat{k}$

Ans :- $\text{div } \vec{F} = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z}$
 $= ye^{xy} + 2\sin y + 2\sin z \cdot \cos z$

$$\begin{aligned} f &= e^{xy} \\ g &= -2\cos y \\ h &= \sin^2 z \end{aligned}$$

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② Determine the constant 'a' so that $(x+3y)\hat{i} + (y-2z)\hat{j} + (x+az)\hat{k}$ is solenoidal.

Ans:-

$\vec{F}$ is solenoidal if $\text{div } \vec{F} = 0$.

$$\Rightarrow \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z} = 0$$

$$\Rightarrow \frac{\partial}{\partial x}(x+3y) + \frac{\partial}{\partial y}(y-2z) + \frac{\partial}{\partial z}(x+az) = 0$$

$$\Rightarrow 1 + 1 + a = 0$$

$$\Rightarrow 2 + a = 0 \Rightarrow \boxed{a = -2}$$

③ Find $\text{div } \vec{F}$ if $\vec{F} = \text{grad}(x^3 + y^3 + z^3 - 3xyz)$
OR find $\nabla \cdot \vec{F}$ if $\vec{F} = \nabla(x^3 + y^3 + z^3 - 3xyz)$

Ans:- Given $\vec{F} = \nabla(x^3 + y^3 + z^3 - 3xyz)$

$$\text{where } \nabla \phi = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$$

$$= \hat{i} \left[\frac{\partial}{\partial x}(x^3 + y^3 + z^3 - 3xyz) \right] + \hat{j} \left[\frac{\partial}{\partial y}(x^3 + y^3 + z^3 - 3xyz) \right] + \hat{k} \left[\frac{\partial}{\partial z}(x^3 + y^3 + z^3 - 3xyz) \right]$$

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$$\therefore \vec{F} = \hat{i}(3x^2 - 3yz) + \hat{j}(3y^2 - 3xz) + \hat{k}(3z^2 - 3xy)$$

$$\begin{aligned}\therefore \operatorname{div} \vec{F} &= \nabla \cdot \vec{F} = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z} \\ &= \frac{\partial}{\partial x}(3x^2 - 3yz) + \frac{\partial}{\partial y}(3y^2 - 3xz) + \frac{\partial}{\partial z}(3z^2 - 3xy) \\ &= 6x + 6y + 6z\end{aligned}$$

④ Prove that $\vec{F} = 3yz\hat{i} + 4xz\hat{j} - 3x^2y\hat{k}$ is solenoidal.

Ans:- $\vec{F}$ is solenoidal if $\operatorname{div} \vec{F} = 0$.

$$\begin{aligned}\operatorname{div} \vec{F} &= \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z} \\ &= \frac{\partial}{\partial x}(3yz) + \frac{\partial}{\partial y}(4xz) - \frac{\partial}{\partial z}(3x^2y) \\ &= 0 + 0 + 0 \\ &= \underline{\underline{0}}\end{aligned}$$

$\therefore \vec{F}$ is solenoidal.

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⑤ Find $\nabla(\nabla \cdot \vec{F})$ at the point $(1, 1, 1)$. Given that $\vec{F} = (3x^2y - z)\hat{i} +$

$$(xz^3 + y^4)\hat{j} - 2x^3z^2\hat{k}$$

Ans :- $\vec{F} = (3x^2y - z)\hat{i} + (xz^3 + y^4)\hat{j} - 2x^3z^2\hat{k}$

$$\nabla \cdot \vec{F} = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z}$$

$$= \frac{\partial}{\partial x}(3x^2y - z) + \frac{\partial}{\partial y}(xz^3 + y^4) + \frac{\partial}{\partial z}(-2x^3z^2)$$

$$= 6xy + 4y^3 - 4x^3z$$

Now $\nabla(\nabla \cdot \vec{F}) = \nabla(6xy + 4y^3 - 4x^3z)$

$$= \hat{i} \frac{\partial}{\partial x}(6xy + 4y^3 - 4x^3z) +$$

$$\hat{j} \frac{\partial}{\partial y}(6xy + 4y^3 - 4x^3z) +$$

$$\hat{k} \frac{\partial}{\partial z}(6xy + 4y^3 - 4x^3z)$$

$$= \hat{i}(6y - 12x^2z) +$$

$$\hat{j}(6x + 12y^2 - 0) +$$

$$\hat{k}(-4x^3)$$

$\therefore$ At $(1, 1, 1)$, $\nabla(\nabla \cdot \vec{F}) = \hat{i}(-6) + \hat{j}(18) + \hat{k}(-4)$

$$= \underline{\underline{-6\hat{i} + 18\hat{j} - 4\hat{k}}}$$

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⑥ If $u = x^2 + y^2 + z^2$, $\vec{v} = x\hat{i} + y\hat{j} + z\hat{k}$
 then show that $\text{div}(u\vec{v}) = 5u$.

Ans:- Given $u = x^2 + y^2 + z^2$
 $\vec{v} = x\hat{i} + y\hat{j} + z\hat{k}$

$$\begin{aligned}\text{Now } u\vec{v} &= (x^2 + y^2 + z^2)(x\hat{i} + y\hat{j} + z\hat{k}) \\ &= (x^3 + y^2x + z^2x)\hat{i} + (x^2y + y^3 + z^2y)\hat{j} \\ &\quad + (x^2z + y^2z + z^3)\hat{k}\end{aligned}$$

$$\text{div}(u\vec{v}) = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z}$$

$$= \frac{\partial}{\partial x} (x^3 + x y^2 + x z^2) +$$

$$\frac{\partial}{\partial y} (x^2 y + y^3 + z^2 y) +$$

$$\frac{\partial}{\partial z} (x^2 z + y^2 z + z^3)$$

$$= (3x^2 + y^2 + z^2) + (x^2 + y^2 + z^2) + (x^2 + y^2 + 3z^2)$$

$$= 5x^2 + 5y^2 + 5z^2$$

$$= 5(x^2 + y^2 + z^2)$$

$$\underline{\underline{= 5u}}$$

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⑦ If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, then
prove that $\nabla \cdot \vec{r} = 3$.

Ans:- $\nabla \cdot \vec{r} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z)$
 $= 1 + 1 + 1 = \underline{\underline{3}}$

⑧ Let $\vec{F} = \vec{F}(x, y, z)$, and $\phi = \phi(x, y, z)$.

Prove that $\text{div}(\phi \vec{F}) = \phi \text{div} \vec{F} + \nabla \phi \cdot \vec{F}$

Ans:- $\phi = \phi(x, y, z)$ is a scalar function

$$\vec{F} = \vec{F}(x, y, z) = f(x, y, z)\hat{i} + g(x, y, z)\hat{j} + h(x, y, z)\hat{k}$$

$$\Rightarrow \phi \vec{F} = \phi f \hat{i} + \phi g \hat{j} + \phi h \hat{k}$$

$$\text{Now } \text{div}(\phi \vec{F}) = \nabla \cdot \phi \vec{F}$$

$$= \frac{\partial}{\partial x}(\phi f) + \frac{\partial}{\partial y}(\phi g) + \frac{\partial}{\partial z}(\phi h)$$

$$= \left[\phi \frac{\partial f}{\partial x} + f \frac{\partial \phi}{\partial x} \right] + \left[\phi \frac{\partial g}{\partial y} + g \frac{\partial \phi}{\partial y} \right]$$

$$+ \left[\phi \frac{\partial h}{\partial z} + h \frac{\partial \phi}{\partial z} \right]$$

$$= \phi \left[\frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z} \right] +$$

$$f \frac{\partial \phi}{\partial x} + g \frac{\partial \phi}{\partial y} + h \frac{\partial \phi}{\partial z}$$

$$= \phi \text{div} \vec{F} + \nabla \phi \cdot \vec{F}$$

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⑨ If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, $r = |\vec{r}|$,

then prove the following:

(a) $\nabla\left(\frac{1}{r}\right) = -\frac{\vec{r}}{r^3}$

(b) $\nabla(r^n) = nr^{n-2}\vec{r}$

(c) $\nabla(f(r)) = \frac{f'(r) \cdot \vec{r}}{r}$

(d) $\nabla \log r = \frac{\vec{r}}{r^2}$

(e) $\nabla^2\left(\frac{1}{r}\right) = 0$

$$\frac{\partial}{\partial x}\left(\frac{1}{r}\right) = \frac{\partial}{\partial r}\left(\frac{1}{r}\right) \cdot \frac{\partial r}{\partial x}$$

(61 a)

① We have $\nabla f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z}$

$$\Rightarrow \nabla(1/r) = \hat{i} \frac{\partial}{\partial x}(1/r) + \hat{j} \frac{\partial}{\partial y}(1/r) + \hat{k} \frac{\partial}{\partial z}(1/r) \quad \text{--- (1)}$$

Now we have $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$$r = |\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

$$\Rightarrow \boxed{r^2 = x^2 + y^2 + z^2} \quad \text{--- (2)}$$

Now differentiating (2) partially w.r.t 'x',

$$\textcircled{2} \Rightarrow \frac{\partial}{\partial x}(r^2) = \frac{\partial}{\partial x}(x^2 + y^2 + z^2)$$

$$\Rightarrow \frac{\partial}{\partial x}(r^2) \cdot \frac{\partial r}{\partial x} = 2x \quad \text{[chain rule]}$$

$$\Rightarrow 2r \cdot \frac{\partial r}{\partial x} = 2x \Rightarrow \boxed{\frac{\partial r}{\partial x} = \frac{x}{r}}$$

Similarly

$$\frac{\partial r}{\partial y} = \frac{y}{r}$$

$$\frac{\partial r}{\partial z} = \frac{z}{r}$$

Now (1) $\Rightarrow$

$$\nabla(1/r) = \hat{i} \left[\frac{\partial}{\partial x}(1/r) \cdot \frac{\partial r}{\partial x} \right] + \hat{j} \left[\frac{\partial}{\partial y}(1/r) \cdot \frac{\partial r}{\partial y} \right] + \hat{k} \left[\frac{\partial}{\partial z}(1/r) \cdot \frac{\partial r}{\partial z} \right]$$

$$= \hat{i} \left[-\frac{1}{r^2} \cdot \frac{x}{r} \right] + \hat{j} \left[-\frac{1}{r^2} \cdot \frac{y}{r} \right] + \hat{k} \left[-\frac{1}{r^2} \cdot \frac{z}{r} \right]$$

$$= -\frac{1}{r^3} [x\hat{i} + y\hat{j} + z\hat{k}]$$

$$= -\frac{1}{r^3} [\vec{r}] = \underline{\underline{-\frac{\vec{r}}{r^3}}}$$

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$$(b) \nabla(r^n) = n r^{n-2} \cdot \vec{r}$$

$$\nabla f = i \frac{\partial f}{\partial x} + j \frac{\partial f}{\partial y} + k \frac{\partial f}{\partial z}$$

$$\nabla(r^n) = i \frac{\partial}{\partial x}(r^n) + j \frac{\partial}{\partial y}(r^n) + k \frac{\partial}{\partial z}(r^n)$$

$$= i \left[n r^{n-1} \frac{\partial r}{\partial x} \right] + j \left[n r^{n-1} \frac{\partial r}{\partial y} \right] + k \left[n r^{n-1} \frac{\partial r}{\partial z} \right]$$

$$= n r^{n-1} \left[i \frac{x}{r} + j \frac{y}{r} + k \frac{z}{r} \right]$$

$$= \frac{n r^{n-1}}{r} [x i + y j + z k]$$

$$= \frac{n r^{n-2}}{r} [x i + y j + z k] = n r^{n-2} \vec{r}$$

$$(c) \nabla f(r) = \frac{f'(r)}{r} \vec{r}$$

$$\nabla f = i \frac{\partial f}{\partial x} + j \frac{\partial f}{\partial y} + k \frac{\partial f}{\partial z}$$

$$\nabla f(r) = i \frac{\partial}{\partial x}(f(r)) + j \frac{\partial}{\partial y}(f(r)) + k \frac{\partial}{\partial z}(f(r))$$

$$= i \left[f'(r) \frac{\partial r}{\partial x} \right] + j \left[f'(r) \frac{\partial r}{\partial y} \right] + k \left[f'(r) \frac{\partial r}{\partial z} \right]$$

$$= f'(r) \left[\frac{x}{r} i + \frac{y}{r} j + \frac{z}{r} k \right]$$

$$= \frac{f'(r)}{r} (x i + y j + z k) = \frac{f'(r)}{r} \vec{r}$$

$$(e) \nabla^2 \left(\frac{1}{r} \right) = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \left(\frac{1}{r} \right)$$

$$= \frac{\partial^2}{\partial x^2} \left(\frac{1}{r} \right) + \frac{\partial^2}{\partial y^2} \left(\frac{1}{r} \right) + \frac{\partial^2}{\partial z^2} \left(\frac{1}{r} \right) \rightarrow (1)$$

$$\text{Take } \frac{\partial^2}{\partial x^2} \left(\frac{1}{r} \right) = \frac{\partial}{\partial x} \left[\frac{\partial}{\partial x} \left(\frac{1}{r} \right) \right] \quad \left[\text{Since } \frac{\partial^2}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} \right) \right]$$

$$= \frac{\partial}{\partial x} \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \right) \cdot \frac{\partial r}{\partial x} \right]$$

$$= \frac{\partial}{\partial x} \left[-\frac{1}{r^2} \cdot \frac{x}{r} \right]$$

$$= \frac{\partial}{\partial x} \left[-\frac{x}{r^3} \right]$$

$$= -\frac{\partial}{\partial x} \left[\frac{x}{r^3} \right]$$

$$= - \left[\frac{r^3 - x \cdot \frac{\partial}{\partial x} (r^3)}{(r^3)^2} \right]$$

By quotient rule

$$= - \left[\frac{r^3 - x \cdot \frac{\partial}{\partial r} (r^3) \cdot \frac{\partial r}{\partial x}}{r^6} \right]$$

$$= - \left[\frac{r^3 - x \cdot 3r^2 \cdot \frac{x}{r}}{r^6} \right]$$

$$= - \left[\frac{r^3 - 3x^2}{r^6} \right]$$

Since $\frac{\partial}{\partial r} \left(\frac{1}{r} \right) = -\frac{1}{r^2}$
and $\frac{\partial r}{\partial x} = \frac{x}{r}$ proved in (a)

Since $\frac{\partial}{\partial r} (r^3) = 3r^2$
& $\frac{\partial r}{\partial x} = \frac{x}{r}$

(64)

$$\therefore \frac{\partial^2}{\partial x^2} \left(\frac{1}{r} \right) = -\frac{1}{r^6} [x^2 - 3x^2 x] \rightarrow (2)$$

$$\text{Similarly, } \frac{\partial^2}{\partial y^2} \left(\frac{1}{r} \right) = -\frac{1}{r^6} [y^2 - 3y^2 y] \rightarrow (3)$$

$$\frac{\partial^2}{\partial z^2} \left(\frac{1}{r} \right) = -\frac{1}{r^6} [z^2 - 3z^2 z] \rightarrow (4)$$

Now substituting (2), (3), (4) in (1),

$$\nabla^2 \left(\frac{1}{r} \right) = -\frac{1}{r^6} [x^2 - 3x^2 x] + \left(-\frac{1}{r^6} \right) [y^2 - 3y^2 y] \\ + \left(-\frac{1}{r^6} \right) [z^2 - 3z^2 z]$$

$$= -\frac{1}{r^6} [x^2 - 3x^2 x + y^2 - 3y^2 y + z^2 - 3z^2 z]$$

$$= -\frac{1}{r^6} [3x^2 - 3x(x^2 + y^2 + z^2)]$$

$$= -\frac{1}{r^6} [3x^2 - 3x \cdot x^2]$$

$$= -\frac{1}{r^6} [3x^2 - 3x^2]$$

$$= \underline{\underline{0}}$$

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$$(d) \nabla(\log r) = \frac{\vec{r}}{r^2}$$

$$\nabla f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z}$$

$$\begin{aligned} \nabla(\log r) &= \hat{i} \frac{\partial}{\partial x} (\log r) + \hat{j} \frac{\partial}{\partial y} (\log r) + \hat{k} \frac{\partial}{\partial z} (\log r) \\ &= \hat{i} \left(\frac{1}{r} \frac{\partial r}{\partial x} \right) + \hat{j} \frac{1}{r} \left(\frac{\partial r}{\partial y} \right) + \hat{k} \frac{1}{r} \frac{\partial r}{\partial z} \end{aligned}$$

$$= \frac{1}{r} \left[\hat{i} \cdot \frac{x}{r} + \hat{j} \frac{y}{r} + \hat{k} \frac{z}{r} \right]$$

$$= \frac{1}{r^2} \left[x \hat{i} + y \hat{j} + z \hat{k} \right]$$

$$= \frac{1}{r^2} (\vec{r}) = \frac{\vec{r}}{r^2}$$

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CURL OF $\vec{F}$

If $\vec{F}(x, y, z) = f(x, y, z)\hat{i} + g(x, y, z)\hat{j} + h(x, y, z)\hat{k}$, then we define the Curl of $\vec{F}$, to be the vector field given by

$$\text{Curl } \vec{F} = \left(\frac{\partial h}{\partial y} - \frac{\partial g}{\partial z} \right) \hat{i} + \left(\frac{\partial f}{\partial z} - \frac{\partial h}{\partial x} \right) \hat{j} + \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) \hat{k}$$

The Curl of $\vec{F}$ can also be represented

as

$$\text{Curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f & g & h \end{vmatrix} \quad \text{which}$$

is the determinant.

* Note :-

Divergence of $\vec{F}$ ($\text{div } \vec{F}$) has scalar values.

Curl of $\vec{F}$ ($\text{Curl } \vec{F}$) has vector values.

Note :- ① Curl relates to the rotational properties of the fluid at a point.

② If $\text{Curl } \vec{F} = 0$, the $\vec{F}$ is said to be irrotational

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① Find $\text{curl } \vec{F}$ if $\vec{F} = e^{xy} \hat{i} - 2 \cos y \hat{j} + \sin z \hat{k}$

Ans:-

$$\text{Curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^{xy} & -2 \cos y & \sin z \end{vmatrix}$$

$$= \hat{i} \left[\frac{\partial}{\partial y} (\sin z) - \frac{\partial}{\partial z} (-2 \cos y) \right] -$$

$$\hat{j} \left[\frac{\partial}{\partial x} (\sin z) - \frac{\partial}{\partial z} (e^{xy}) \right] +$$

$$\hat{k} \left[\frac{\partial}{\partial x} (-2 \cos y) - \frac{\partial}{\partial y} (e^{xy}) \right]$$

$$= \hat{i} [0 - 0] - \hat{j} [0 - 0] + \hat{k} [0 - x e^{xy}]$$

$$= \underline{\underline{-x e^{xy} \hat{k}}}$$

② Find the constants a, b, c if

$$\vec{F} = (3x - 3y + az) \hat{i} + (bx + 2y - 4z) \hat{j} + (2x + cy + z) \hat{k} \text{ is irrotational.}$$

Ans:-

$\vec{F}$ is irrotational if $\text{curl } \vec{F} = 0$

$$\text{i.e. } \nabla \times \vec{F} = 0$$

Field

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x-3y+az & bx+ay-4z & 2x+cy+z \end{vmatrix} = 0$$

$$= \hat{i} \left[\frac{\partial}{\partial y} (2x+cy+z) - \frac{\partial}{\partial z} (bx+ay-4z) \right] \\ - \hat{j} \left[\frac{\partial}{\partial x} (2x+cy+z) - \frac{\partial}{\partial z} (3x-3y+az) \right] \\ + \hat{k} \left[\frac{\partial}{\partial x} (bx+ay-4z) - \frac{\partial}{\partial y} (3x-3y+az) \right]$$

$$= 0$$

$$\Rightarrow \hat{i} [c+4] - \hat{j} [2-a] + \hat{k} [b+3] = 0$$

$$\Rightarrow c+4=0, \quad -(2-a)=0, \quad b+3=0$$

$$\Rightarrow \underline{c=-4}, \quad \underline{a=2}, \quad \underline{b=-3}$$

③ If $\vec{V} = \vec{W} \times \vec{r}$, then prove that

$W = \frac{1}{2} \text{Curl} \cdot \vec{V}$, where W is a constant vector and $\vec{r}$ is the position vector.

Ans:-

$$\text{We have } \vec{V} = \vec{W} \times \vec{r}$$

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let $\vec{W} = W_1 \hat{i} + W_2 \hat{j} + W_3 \hat{k}$ and

$\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$

$$\vec{V} = \vec{W} \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ W_1 & W_2 & W_3 \\ x & y & z \end{vmatrix}$$

$$= \hat{i} [W_2 z - W_3 y] - \hat{j} [W_1 z - W_3 x] + \hat{k} [W_1 y - W_2 x]$$

$$\text{Curl } \vec{V} = \nabla \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ W_2 z - W_3 y & -(W_1 z - W_3 x) & W_1 y - W_2 x \end{vmatrix}$$

$$= \hat{i} \left[\frac{\partial}{\partial y} (W_1 y - W_2 x) - \frac{\partial}{\partial z} (-W_1 z + W_3 x) \right] -$$

$$\hat{j} \left[\frac{\partial}{\partial x} (W_1 y - W_2 x) - \frac{\partial}{\partial z} (W_2 z - W_3 y) \right] +$$

$$\hat{k} \left[\frac{\partial}{\partial x} (W_1 z + W_3 x) - \frac{\partial}{\partial y} (W_2 z - W_3 y) \right]$$

$$= \hat{i} [W_1 + W_1] - \hat{j} [-W_2 - W_2] + \hat{k} [W_3 + W_3]$$

$$= \hat{i} [2W_1] - \hat{j} [-2W_2] + \hat{k} [2W_3]$$

$$= 2 [W_1 \hat{i} + W_2 \hat{j} + W_3 \hat{k}] = 2 \vec{W}$$

$$\Rightarrow \vec{W} = \frac{1}{2} \text{Curl } \vec{V}$$

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④ If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, find $\nabla \times \vec{r}$.

Ans:- Given $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.

$$\begin{aligned}\nabla \times \vec{r} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} \\ &= \hat{i} \left[\frac{\partial}{\partial y}(z) - \frac{\partial}{\partial z}(y) \right] - \hat{j} \left[\frac{\partial}{\partial x}(z) - \frac{\partial}{\partial z}(x) \right] \\ &\quad + \hat{k} \left[\frac{\partial}{\partial x}(y) - \frac{\partial}{\partial y}(x) \right] \\ &= \hat{i} [0 - 0] - \hat{j} [0 - 0] + \hat{k} [0 - 0] \\ &= \underline{\underline{0}}\end{aligned}$$

⑤ Prove that $\text{curl}(\text{grad } \phi) = 0$ OR

Prove that $\nabla \times (\nabla \phi) = 0$.

Ans:- $\nabla \phi = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$

$$\begin{aligned}\nabla \times \nabla \phi &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial z} \end{vmatrix} \\ &= \hat{i} \left[\frac{\partial}{\partial y} \left(\frac{\partial \phi}{\partial z} \right) - \frac{\partial}{\partial z} \left(\frac{\partial \phi}{\partial y} \right) \right] - \hat{j} \left[\frac{\partial}{\partial x} \left(\frac{\partial \phi}{\partial z} \right) - \frac{\partial}{\partial z} \left(\frac{\partial \phi}{\partial x} \right) \right] \\ &\quad + \hat{k} \left[\frac{\partial}{\partial x} \left(\frac{\partial \phi}{\partial y} \right) - \frac{\partial}{\partial y} \left(\frac{\partial \phi}{\partial x} \right) \right]\end{aligned}$$

$$\begin{aligned}
 &= \hat{i} \left[\frac{\partial^2 \phi}{\partial y \partial z} - \frac{\partial^2 \phi}{\partial z \partial y} \right] + \hat{j} \left[-\frac{\partial^2 \phi}{\partial x \partial z} + \frac{\partial^2 \phi}{\partial z \partial x} \right] \\
 &\quad + \hat{k} \left[\frac{\partial^2 \phi}{\partial x \partial y} - \frac{\partial^2 \phi}{\partial y \partial x} \right] \\
 &= \underline{\underline{0}}
 \end{aligned}$$

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⑥ Prove that $\text{div}(\text{curl } \vec{F}) = 0$ OR

Prove that $\nabla \cdot (\nabla \times \vec{F}) = 0$.

⑦ Show that $\vec{F} = (x \cos x + \sin y) \hat{i} + x(y \cos z + \sin z) \hat{j} + (y \cos z + \sin x) \hat{k}$ is irrotational.

Ans:-

⑥ Let $\vec{F} = f \hat{i} + g \hat{j} + h \hat{k}$

$$\begin{aligned}
 \nabla \times \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f & g & h \end{vmatrix} \\
 &= \hat{i} \left[\frac{\partial h}{\partial y} - \frac{\partial g}{\partial z} \right] - \hat{j} \left[\frac{\partial h}{\partial x} - \frac{\partial f}{\partial z} \right] \\
 &\quad + \hat{k} \left[\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right]
 \end{aligned}$$

(7)

$$\begin{aligned} \text{Now } \nabla \cdot (\nabla \times \vec{F}) &= \frac{\partial}{\partial x} \left(\frac{\partial h}{\partial y} - \frac{\partial g}{\partial z} \right) + \\ &\quad \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial z} - \frac{\partial h}{\partial x} \right) + \\ &\quad \frac{\partial}{\partial z} \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) \\ &= 0 + 0 + 0 = \underline{\underline{0}} \end{aligned}$$

$$\therefore \underline{\underline{\nabla \cdot (\nabla \times \vec{F}) = 0}}$$

⑦ We have to show that $\text{Curl } \vec{F} = 0$

$$\text{or } \nabla \times \vec{F} = 0$$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z \cos x + \sin y & x \cos y + \sin z & y \cos z + \sin x \end{vmatrix}$$

$$= \hat{i} \left[\frac{\partial}{\partial y} (y \cos z + \sin x) - \frac{\partial}{\partial z} (x \cos y + \sin z) \right]$$

$$- \hat{j} \left[\frac{\partial}{\partial x} (y \cos z + \sin x) - \frac{\partial}{\partial z} (z \cos x + \sin y) \right]$$

$$+ \hat{k} \left[\frac{\partial}{\partial x} (x \cos y + \sin z) - \frac{\partial}{\partial y} (z \cos x + \sin y) \right]$$

$$= \hat{i} [\cos z - \cos z] - \hat{j} [\cos x - \cos x]$$

$$+ \hat{k} [\cos y - \cos y] = 0$$

Hence $\vec{F}$ is irrotational

(73)

Problems

- ① Show that the vector $\vec{F} = (2xy + z^3)\hat{i} + (x^2\hat{j} + 3xz^2)\hat{k}$ is irrotational. Also find the corresponding scalar function.

Ans: If $\vec{F}$ is irrotational, then

$$\nabla \times \vec{F} = 0.$$

$$\text{Now } \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy + z^3 & x^2 & 3xz^2 \end{vmatrix}$$

$$= \hat{i} \left[\frac{\partial}{\partial y} (3xz^2) - \frac{\partial}{\partial z} (x^2) \right] +$$

$$\hat{j} \left[\frac{\partial}{\partial x} (3xz^2) - \frac{\partial}{\partial z} (2xy + z^3) \right] +$$

$$\hat{k} \left[\frac{\partial}{\partial x} (x^2) - \frac{\partial}{\partial y} (2xy + z^3) \right]$$

$$= \hat{i} [0 - 0] - \hat{j} [3z^2 - 3z^2] + \hat{k} [2x - 2x]$$

$$= \underline{\underline{0}}$$

The corresponding scalar function is

$$\vec{F} = \nabla \phi.$$

$$(2xy + z^3)\hat{i} + x^2\hat{j} + 3xz^2\hat{k} = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$$

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Equating coefficients of i, j, k components

$$\frac{\partial \phi}{\partial x} = 2xy + z^3 \rightarrow \textcircled{1}$$

$$\frac{\partial \phi}{\partial y} = x^2 \rightarrow \textcircled{2}$$

$$\frac{\partial \phi}{\partial z} = 3xz^2 \rightarrow \textcircled{3}$$

Integrating $\textcircled{1}$ w.r.t ' x ', we have

$$\phi = \frac{2x^2y}{2} + z^3x + \text{terms without 'x'}$$

Integrating $\textcircled{2}$ w.r.t ' y ',

$$\phi = x^2y + \text{terms without 'y'}$$

Integrating $\textcircled{3}$ w.r.t ' z ',

$$\phi = 3x \frac{z^3}{3} + \text{terms without 'z'}$$

Hence $\phi = x^2y + z^3x + k$ which is

the corresponding scalar function.

$$\frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} = x^2 \hat{i} + (2xy + z^3) \hat{j} + 3xz^2 \hat{k}$$

Conservative Fields and Potential Functions.

Definition :-

A vector field $\vec{F}$ in 2-space or 3-space is said to be conservative in a region if it is the gradient field for some function ϕ in that region, that is if

$$\boxed{\vec{F} = \nabla \phi.}$$

The function ϕ is called a potential function for $\vec{F}$ in the region. and $-\phi$ is the potential energy.

eg:- Find $\nabla \cdot \vec{F}$ whose potential function is $\phi = xy + yz + zx$.

Ans:-

If ϕ is the potential function, then the corresponding vector is $\nabla \phi$.

$$\text{ie } \vec{F} = \nabla \phi.$$

$$= \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$$

$$= \hat{i}(y+z) + \hat{j}(x+z) + \hat{k}(y+x)$$

(76)

eg:- Prove that $\phi(x, y) = \tan^{-1}(xy)$ is the potential function for the vector

$$\vec{F} = \left(\frac{y}{1+x^2+y^2} \right) \hat{i} + \left(\frac{x}{1+x^2+y^2} \right) \hat{j}$$

Ans:- Given $\phi(x, y) = \tan^{-1}(xy)$

We have $\vec{F} = \nabla \phi$.

$$\nabla \phi = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$$

$$= \hat{i} \left[\frac{1}{1+x^2+y^2} \cdot y \right] + \hat{j} \left[\frac{1}{1+x^2+y^2} \cdot x \right]$$

$$= \left(\frac{y}{1+x^2+y^2} \right) \hat{i} + \left(\frac{x}{1+x^2+y^2} \right) \hat{j}$$

$$= \vec{F}$$

$$\underline{\underline{\quad}}$$

(77)

② Prove that the vector $\vec{F} = (x+yz)\hat{i} + (4y+zx)\hat{j} + (6z-xy)\hat{k}$ forms a conservative field. Also find the corresponding potential function and potential energy.

Ans :- To show that $\vec{F}$ forms a conservative field, it is enough to show that $\nabla \times \vec{F} = 0$.

$$\begin{aligned}\nabla \times \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x+yz & 4y+zx & 6z-xy \end{vmatrix} \\ &= \hat{i} \left[\frac{\partial}{\partial y}(xy-6z) - \frac{\partial}{\partial z}(xy+zx) \right] - \\ &\quad \hat{j} \left[\frac{\partial}{\partial x}(xy-6z) - \frac{\partial}{\partial z}(2x+yz) \right] + \\ &\quad \hat{k} \left[\frac{\partial}{\partial x}(4y+zx) - \frac{\partial}{\partial y}(2x+yz) \right] \\ &= \hat{i} [x-x] - \hat{j} [y-y] + \hat{k} [z-z] \\ &= \underline{\underline{0}}\end{aligned}$$

Hence $\vec{F}$ forms a conservative field.

Corresponding potential function is given by $\vec{F} = \nabla \phi$.

$$(2x + yz)\hat{i} + (4y + zx)\hat{j} - (6z - xy)\hat{k} \\ = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$$

$$\Rightarrow \frac{\partial \phi}{\partial x} = 2x + yz \Rightarrow \phi = \frac{2x^2}{2} + yzx + \text{terms without } x$$

$$\frac{\partial \phi}{\partial y} = 4y + zx \Rightarrow \phi = \frac{4y^2}{2} + xyz + \text{terms without } y$$

$$\frac{\partial \phi}{\partial z} = xy - 6z \Rightarrow \phi = xyz - \frac{6z^2}{2} + \text{terms without } z$$

$$\therefore \phi = \underline{xyz + x^2 + 2y^2 - 3z^2 + k}$$

$$\text{Potential energy} = -\phi$$

$$= -(xyz + x^2 + 2y^2 - 3z^2 + k) = \underline{3z^2 - xyz - x^2 - 2y^2 - k}$$

(+9)

LINE INTEGRALS

If C is a smooth curve parametrized by $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j}$, $a \leq t \leq b$, then

$$\boxed{\int_C f(x, y) ds = \int_a^b f(x(t), y(t)) \|\vec{r}'(t)\| dt} \rightarrow \textcircled{1}$$

Similarly,

If C is a smooth curve in 3-space parametrized by $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$, $a \leq t \leq b$, then

$$\boxed{\int_C f(x, y, z) ds = \int_a^b f(x(t), y(t), z(t)) \|\vec{r}'(t)\| dt} \rightarrow \textcircled{2}$$

(80)

① Evaluate $\int_C (1+xy^2) ds$ where C is given by $C: \vec{r}(t) = t\hat{i} + 2t\hat{j}$ ($0 \leq t \leq 1$)

Ans:-

We have $\int_C f(x,y) ds = \int_a^b f(x(t), y(t)) \|\vec{r}'(t)\| dt$

Now $\vec{r}(t) = t\hat{i} + 2t\hat{j} \Rightarrow \boxed{x(t)=t, y(t)=2t}$

$\Rightarrow \vec{r}'(t) = \hat{i} + 2\hat{j} \Rightarrow \|\vec{r}'(t)\| = \sqrt{1+2^2} = \underline{\underline{\sqrt{5}}}$

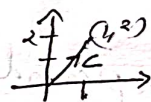
$\therefore \int_C f(x,y) ds = \int_a^b f(x(t), y(t)) \|\vec{r}'(t)\| dt$

$\int_C (1+xy^2) ds = \int_0^1 [1+t(2t)^2] \sqrt{5} dt$

$= \sqrt{5} \int_0^1 (1+4t^3) dt$

$= \sqrt{5} \left[t + \frac{4t^4}{4} \right]_0^1 = \sqrt{5} [1+1-0]$

$= \underline{\underline{2\sqrt{5}}}$

$f(x,y) = 1+xy^2$
 $f(t, 2t) = 1+t(2t)^2$
 and
 $0 \leq t \leq 1 \Rightarrow a=0, b=1$


② Evaluate $\int_C \frac{1}{1+x} ds$ where $C: \vec{r}(t) = t\hat{i} + \frac{2}{3}t^{3/2}\hat{j}$, $0 \leq t \leq 4$

Ans:-

Given that $\vec{r}(t) = t\hat{i} + \frac{2}{3}t^{3/2}\hat{j} \Rightarrow x(t) = t$
 $y(t) = \frac{2}{3}t^{3/2}$

(81)

$$\Rightarrow \vec{r}'(t) = \hat{i} + \frac{2}{3} \cdot \frac{3}{2} t^{1/2} \hat{j}$$

$$= \hat{i} + t^{1/2} \hat{j} = \hat{i} + \sqrt{t} \hat{j}$$

$$\Rightarrow \|\vec{r}'(t)\| = \sqrt{1^2 + (\sqrt{t})^2} = \sqrt{1+t}$$

$$\therefore \int_C f(x, y) ds = \int_a^b f(x(t), y(t)) \|\vec{r}'(t)\| dt$$

$$\int_C \frac{1}{1+x} ds = \int_0^4 f\left(t, \frac{2}{3} t^{3/2}\right) \sqrt{1+t} dt$$

$$= \int_0^4 \frac{1}{1+t} \sqrt{1+t} dt$$

$$= \int_0^4 \frac{1}{\sqrt{1+t}} dt = \int_0^4 (1+t)^{-1/2} dt$$

$$= \left[\frac{(1+t)^{-1/2+1}}{-1/2+1} \right]_0^4 = \left[\frac{(1+t)^{1/2}}{1/2} \right]_0^4$$

$$= 2 \left[5^{1/2} - 1 \right] = \underline{\underline{2[\sqrt{5} - 1]}}$$

③ Evaluate the line integral $\int_C (xy + z^3) ds$ from $(1, 0, 0)$ to $(-1, 0, \pi)$ along the helix C that is represented by the parametric equations $x = \cos t$, $y = \sin t$, $z = t$ where $0 \leq t \leq \pi$

(82)

Ans: Given $x(t) = \cos t$, $y(t) = \sin t$, $z(t) = t$

$$\therefore \vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$$

$$\Rightarrow \vec{r}(t) = (\cos t)\hat{i} + (\sin t)\hat{j} + t\hat{k}$$

$$\Rightarrow \vec{r}'(t) = (-\sin t)\hat{i} + (\cos t)\hat{j} + \hat{k}$$

$$\Rightarrow \|\vec{r}'(t)\| = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1}$$

$$= \sqrt{\sin^2 t + \cos^2 t + 1} = \underline{\underline{\sqrt{2}}}$$

$$\int_C f(x, y, z) ds = \int_a^b f(x(t), y(t), z(t)) \|\vec{r}'(t)\| dt$$

$$\int_C (xy + z^3) ds = \int_0^{\pi} [(\cos t \cdot \sin t) + t^3] \sqrt{2} dt$$

$$= \sqrt{2} \int_0^{\pi} \frac{\sin 2t}{2} + t^3 dt$$

$$\boxed{\sin 2t = 2 \sin t \cos t}$$

$$= \sqrt{2} \left[-\frac{\cos 2t}{4} + \frac{t^4}{4} \right]_0^{\pi}$$

$$= \sqrt{2} \left[\left(-\frac{\cos 2\pi}{4} + \frac{\pi^4}{4} \right) - \left(-\frac{1}{4} + 0 \right) \right]$$

$$= \sqrt{2} \left[-\frac{1}{4} + \frac{\pi^4}{4} + \frac{1}{4} \right]$$

$$= \underline{\underline{\frac{\sqrt{2} \pi^4}{4}}}$$

④ Suppose that a semicircular wire has the equation $y = \sqrt{25 - x^2}$ and that its ~~linear~~ density is $\delta(x, y) = 15 - y$. Find the mass of the wire.

Ans :-

Note:- If C is a curve in 2-space that represents a thin wire and if $f(x, y)$ is the linear density function of the wire, then the mass of the wire is given by $M = \int_C f(x, y) ds$.

$$\begin{aligned} \text{Here } y &= \sqrt{25 - x^2} \Rightarrow y^2 = 25 - x^2 \\ &\Rightarrow x^2 + y^2 = 25, \text{ circle} \\ &\Rightarrow x(t) = r \cos t = 5 \cos t \\ &\quad y(t) = r \sin t = 5 \sin t. \end{aligned}$$

$$\begin{aligned} \therefore \vec{r}(t) &= x(t)\hat{i} + y(t)\hat{j} \\ &= (5 \cos t)\hat{i} + (5 \sin t)\hat{j} \\ \vec{r}'(t) &= (-5 \sin t)\hat{i} + (5 \cos t)\hat{j} \end{aligned}$$

$$\|\vec{r}'(t)\| = \sqrt{(-5 \sin t)^2 + (5 \cos t)^2} = \sqrt{25} = 5$$

(84)

$$M = \int_C f(x, y) ds$$

$$= \int_C (15 - y) ds$$

$$= \int_C (15 - y(t)) \|\vec{r}'(t)\| dt, \quad 0 \leq t \leq \pi$$

$$= \int_0^\pi (15 - 5 \sin t) 5 dt$$

$$= 5 [15t + 5 \cos t]_0^\pi$$

$$= 5 [(15\pi + 5 \cos \pi) - (0 + 5)]$$

$$= 5 [15\pi + 5(-1) - 5] = 5 [15\pi - 10]$$

$$= \underline{\underline{75\pi - 50}}$$

⑤ Evaluate $\int_C y dx - x dz + z dy$ along $x=t, y=t^2, z=t^3$ from $(0,0,0)$ to $(1,1,1)$

Ans :- $x=t \Rightarrow dx=dt$, $y=t^2 \Rightarrow dy=2t dt$
 $z=t^3 \Rightarrow dz=3t^2 dt$

$$\begin{aligned} \therefore \int_C y dx - x dz + z dy &= \int_0^1 t^2 \cdot dt + \int_0^1 t \cdot (3t^2 dt) + \int_0^1 t^3 \cdot (2t) dt \\ &= \left[\frac{t^3}{3} - \frac{3t^4}{4} + \frac{2t^5}{5} \right]_0^1 = \frac{1}{3} + \frac{3}{4} + \frac{2}{5} \\ &= \underline{\underline{\frac{-1}{60}}} // \end{aligned}$$

Line Integrals of Vector Fields.

* Definition:-

If $\vec{F}$ is a continuous vector field and C is a smooth oriented curve, then the line integral of $\vec{F}$ along C

is $\boxed{\int_C \vec{F} \cdot d\vec{r}}$

where (a) $\vec{F} = f(x, y) \hat{i} + g(x, y) \hat{j}$ and

$$d\vec{r} = dx \hat{i} + dy \hat{j} \text{ in 2-space}$$

and (b) $\vec{F} = f(x, y, z) \hat{i} + g(x, y, z) \hat{j} + h(x, y, z) \hat{k}$

$$d\vec{r} = dx \hat{i} + dy \hat{j} + dz \hat{k} \text{ in 3-space.}$$

Q. 86a

① Evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F}(x, y) = \cos x \hat{i} + \sin x \hat{j}$

where C is (a) $\vec{r}(t) = -\frac{\pi}{2} \hat{i} + t \hat{j}$ ($1 \leq t \leq 2$)

(b) $\vec{r}(t) = t \hat{i} + t^2 \hat{j}$ ($-1 \leq t \leq 2$)

Ans:-

$$\textcircled{a} \int_C \vec{F} \cdot d\vec{r} = \int_C (\cos x \hat{i} + \sin x \hat{j}) \cdot (dx \hat{i} + dy \hat{j})$$

$$= \int_C \cos x dx + \sin x dy \quad \left| \begin{array}{l} \vec{r}(t) = -\frac{\pi}{2} \hat{i} + t \hat{j} \\ \Rightarrow x(t) = -\frac{\pi}{2} \\ y(t) = t \end{array} \right.$$

$$= \int_C \cos(x(t)) dx + \sin(x(t)) dy$$

$$= \int_1^2 \cos\left(-\frac{\pi}{2}\right) (0 dt) + \sin\left(-\frac{\pi}{2}\right) (1 dt)$$

and $1 \leq t \leq 2$.

$$= \int_1^2 0 dt + (-1) dt$$

$$= [-t]_1^2 = -2 + 1 = \underline{\underline{-1}}$$

② From (a) we have

$$\int_C \vec{F} \cdot d\vec{r} = \int_C \cos(x(t)) dx + \sin(x(t)) dy$$

$$= \int_{-1}^2 \cos t dt + \sin t (2t dt)$$

$$= \int_{-1}^2 \cos t dt + \int_{-1}^2 2t \sin t dt$$

$$\left| \begin{array}{l} \vec{r}(t) = t \hat{i} + t^2 \hat{j} \\ \Rightarrow x(t) = t \\ y(t) = t^2 \end{array} \right.$$

$$\Rightarrow x(t) = t$$

$$y(t) = t^2$$

$$\Rightarrow dx = 1 dt$$

$$dy = 2t dt$$

$t: -1 \text{ to } 2$

$$= [\sin t]_{-1}^2 + [(2t)(-\cos t) - (2)(-\sin t)]_{-1}^2$$

$$= [\sin 2 - \sin(-1)] + [(2 \times 2)(-\cos 2) + 2 \sin 2 - (-2)(-\cos(-1)) + 2 \sin(-1)]$$

$$= \sin 2 + \sin 1 - 4 \cos 2 + 2 \sin 2 + (-2) \cos 1 + 2 \sin 1$$

$$\begin{aligned} \cos(-\theta) &= \cos \theta \\ \sin(-\theta) &= -\sin \theta \end{aligned}$$

$$= 3 \sin 2 - 4 \cos 2 + 3 \sin 1 - 2 \cos 1$$

$$= 3 \sin 2 - 4 \cos 2 + 3 \sin 1 - 2 \cos 1$$

HW Evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F}(x, y) = -y \hat{i} + x \hat{j}$
 and C is $\vec{r}(t) = t \hat{i} + 2t \hat{j}$ ($0 \leq t \leq 1$)

Ans:

$$\int_C \vec{F} \cdot d\vec{r} = \int_C (-y \hat{i} + x \hat{j}) \cdot (dx \hat{i} + dy \hat{j})$$

$$= \int_C -y dx + x dy$$

$$= \int_0^1 -2t dt + t(2 dt)$$

$$= -2 \left[\frac{t^2}{2} \right]_0^1 + 2 \left[\frac{t^2}{2} \right]_0^1$$

$$= -2 \left[\frac{1}{2} - 0 \right] + 2 \left[\frac{1}{2} - 0 \right] = \underline{\underline{0}}$$

$$\begin{aligned} \vec{r}(t) &= t \hat{i} + 2t \hat{j} \\ \Rightarrow x(t) &= t \Rightarrow dx = 1 dt \\ y(t) &= 2t \Rightarrow dy = 2 dt \\ \text{Also } (0 \leq t \leq 1) \end{aligned}$$

(88)

Work as a Line Integral

Definition :-

Suppose that under the influence of a continuous force field $\vec{F}$, a particle moves along a smooth curve C and C is oriented in the direction of motion of the particle. Then the work performed by the force field on the particle is

$$\int_C \vec{F} \cdot d\vec{r}$$

- ① Find the work done by the force field $\vec{F} = xy\hat{i} + x^2\hat{j}$ on a particle that moves along the curve $C: x=y^2$ from $(0,0)$ to $(1,1)$

Ans:- We have $W = \int_C \vec{F} \cdot d\vec{r}$

$$= \int_C (xy\hat{i} + x^2\hat{j}) \cdot (dx\hat{i} + dy\hat{j})$$

$$= \int_C xy dx + 0 + 0 + x^2 dy$$

$$= \int_C xy dx + x^2 dy$$

$\begin{aligned} \vec{r} &= x\hat{i} + y\hat{j} \\ d\vec{r} &= dx\hat{i} + dy\hat{j} \end{aligned}$

$\begin{aligned} C: x &= y^2 \\ \Rightarrow dx &= 2y dy \end{aligned}$

(89)

$$= \int_0^1 y^2 \cdot y \cdot (2y \, dy) + (y^2)^3 \, dy \quad \left| \begin{array}{l} (0,0) \text{ to } (1,1) \\ \Rightarrow y \rightarrow 0 \text{ to } 1 \end{array} \right.$$

$$= \int_0^1 2y^4 \, dy + y^6 \, dy$$

$$= \left[2 \frac{y^5}{5} + \frac{y^7}{7} \right]_0^1 = \frac{2}{5} + \frac{1}{7} = \underline{\underline{\frac{19}{35}}}$$

② Find the work done by the force $\vec{F} = xy \hat{i} + yz \hat{j} + xz \hat{k}$ on a particle that moves along the curve C is given by $\vec{r}(t) = t\hat{i} + t^2\hat{j} + t^3\hat{k}$, $0 \leq t \leq 1$

Ans:- We have $W = \int_C \vec{F} \cdot d\vec{r}$ $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$
 $d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$

$$= \int_C [(xy)\hat{i} + (yz)\hat{j} + (xz)\hat{k}] \cdot [dx\hat{i} + dy\hat{j} + dz\hat{k}]$$

$$= \int_C xy \, dx + yz \, dy + xz \, dz$$

$$= \int_0^1 t \cdot t^2 \, dt + t^2 \cdot t^3 \cdot 2t \, dt + t \cdot t^3 \cdot 3t^2 \, dt$$

$$= \int_0^1 t^3 \, dt + 2t^6 \, dt + 3t^6 \, dt$$

$$= \int_0^1 (t^3 + 5t^6) \, dt = \left[\frac{t^4}{4} + \frac{5t^7}{7} \right]_0^1 = \frac{1}{4} + \frac{5}{7}$$

$$= \frac{27}{28} //$$

$$\vec{r}(t) = t\hat{i} + t^2\hat{j} + t^3\hat{k}$$

$$\Rightarrow x(t) = t \Rightarrow dx = dt$$

$$y(t) = t^2 \Rightarrow dy = 2t \, dt$$

$$z(t) = t^3 \Rightarrow dz = 3t^2 \, dt$$

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③ Evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \sin x \hat{i} + \cos x \hat{j}$
 where C is the curve $C: \vec{r}(t) = \pi t \hat{i} + t \hat{j}$, $0 \leq t \leq 2$

Ans:-

$$\begin{aligned} \int_C \vec{F} \cdot d\vec{r} &= \int_C (\sin x \hat{i} + \cos x \hat{j}) (dx \hat{i} + dy \hat{j}) \left| \begin{array}{l} \vec{r}(t) = \pi t \hat{i} + t \hat{j} \\ \Rightarrow x = \pi t \Rightarrow dx = \pi dt \\ y = t \Rightarrow dy = dt \end{array} \right. \\ &= \int_0^2 (\sin \pi t \cdot \pi dt + \cos \pi t \cdot dt) \\ &= \int_0^2 \sin \pi t \cdot \pi dt + \cos \pi t dt \\ &\Rightarrow \int_0^2 0 + (-1) dt = [-t]_0^2 = \underline{\underline{-2}} \end{aligned}$$

④ Find the work done by the force
 $\vec{F} = (x^2 + xy) \hat{i} + (y - x^2y) \hat{j}$ on a particle
 that moves along $C: x=t, y=\frac{1}{t}, 1 \leq t \leq 3$.

Ans:-

$$\begin{aligned} \int_C \vec{F} \cdot d\vec{r} &= \int_C [(x^2 + xy) \hat{i} + (y - x^2y) \hat{j}] [dx \hat{i} + dy \hat{j}] \\ &= \int_C (x^2 + xy) dx + (y - x^2y) dy \left| \begin{array}{l} x=t \\ y=\frac{1}{t} \\ dx=dt \\ dy=-\frac{1}{t^2} dt \end{array} \right. \\ &= \int_1^3 (t^2 + t \cdot \frac{1}{t}) dt + (\frac{1}{t} - t^2 \cdot \frac{1}{t}) (-\frac{1}{t^2} dt) \end{aligned}$$

$$\begin{aligned}
 &= \int_1^3 (t^2 + 1) dt + \left(-\frac{1}{t^3} + \frac{1}{t} \right) dt \\
 &= \left[\frac{t^3}{3} + t - \frac{t^{-3}}{-3} + \log t \right]_1^3 \\
 &= \left[\frac{3^3}{3} + 3 + \frac{3^{-3}}{3} + \log 3 \right] - \left[\frac{1}{3} + 1 + \frac{1^{-3}}{3} + \log 1 \right] \\
 &= \left(12 + \frac{1}{18} + \log 3 \right) - \left(\frac{4}{3} + \frac{1}{2} + 0 \right) \\
 &= 12 + \frac{1}{18} - \frac{11}{6} + \log 3 = \underline{\underline{10.69}}
 \end{aligned}$$

⑤ Evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = y\hat{i} - x\hat{j}$ along the triangle joining the vertices $(0,0)$, $(1,0)$ and $(0,1)$

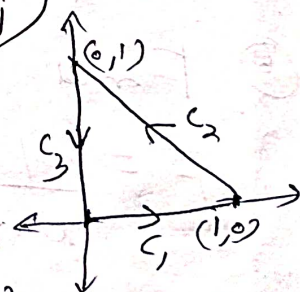
Ans:-

$$\int_C \vec{F} \cdot d\vec{r} = \int_C (y\hat{i} - x\hat{j}) (dx\hat{i} + dy\hat{j})$$

$$= \int_C y dx - x dy$$

$$= \int_{C_1} (y dx - x dy) + \int_{C_2} (y dx - x dy)$$

$$+ \int_{C_3} (y dx - x dy) \longrightarrow \textcircled{1}$$



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$C_1: y=0 \Rightarrow \boxed{dy=0}$ and $x:$

$C_2: \text{line joining } (1,0) \text{ and } (0,1)$

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} \Rightarrow \frac{x-1}{0-1} = \frac{y-0}{1-0}$$

$$\Rightarrow x-1 = -y$$

$$\Rightarrow x = 1 - y, \quad y: 0 \rightarrow 1$$

$$\Rightarrow \boxed{dx = -dy}$$

$C_3: x=0 \Rightarrow \boxed{dx=0}$

$$\textcircled{1} \Rightarrow \int_C \vec{F} \cdot d\vec{r} = \int_{C_1} 0 - 0 + \int_{C_2} y dy - (1+y) dy$$

$$= \int_0^1 (y-1-y) dy = \int_0^1 -1 dy$$

$$= [-y]_0^1 = \underline{\underline{-1}}$$

⑥ Find the work done by the force

$$\vec{F} = \frac{1}{x^2+y^2} \hat{i} + \frac{y}{x^2+y^2} \hat{j} \text{ for a particle that}$$

moves along the curve C where C is the portion of the circle $x^2+y^2=16$ which lies in the first quadrant.

Ans:- $W = \int_C \vec{F} \cdot d\vec{r}$.

$$= \int_C \left[\left(\frac{1}{x^2+y^2} \right) i + \left(\frac{4}{x^2+y^2} \right) j \right] [dx i + dy j]$$

$$= \int_C \frac{1}{x^2+y^2} dx + \frac{4}{x^2+y^2} dy$$

$$= \int_0^{\pi/2} \frac{-4 \sin \omega d\omega}{16} + \frac{4 \cdot 4 \cos \omega d\omega}{16}$$

$$= \int_0^{\pi/2} \left[-\frac{\sin \omega d\omega}{4} + \cos \omega d\omega \right]$$

$$= \left[\frac{\cos \omega}{4} + \sin \omega \right]_0^{\pi/2}$$

$$= \left(\frac{0}{4} + 1 \right) - \left(\frac{1}{4} + 0 \right)$$

$$= 1 - \frac{1}{4} = \frac{3}{4}$$

$$\begin{aligned} x^2+y^2 &= 16 \\ x &= 4 \cos \omega \\ y &= 4 \sin \omega \\ \text{when } r &= 4 \\ \text{and } \omega &: 0 \text{ to } 2\pi \\ \Rightarrow \\ dx &= -4 \sin \omega d\omega \\ dy &= 4 \cos \omega d\omega \end{aligned}$$

As it is in the 1st quadrant, $\omega \rightarrow 0 \text{ to } \pi/2$

⑦ Evaluate $\int_C y dx + z dy + x dz$ along the line joining $(0,0,0)$ to $(1,1,1)$.

Ans:-

Equation of line joining $(0,0,0)$ and $(1,1,1)$

is $\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$

$$\Rightarrow \frac{x-0}{1-0} = \frac{y-0}{1-0} = \frac{z-0}{1-0}$$

$$\Rightarrow \boxed{x=y=z}$$

$$\begin{aligned} \therefore \int_C y dx + z dy + x dz &= \int_0^1 x dx + x dx + x dx \\ &= \int_0^1 3x dx = \left[\frac{3x^2}{2} \right]_0^1 \\ &= \underline{\underline{\frac{3}{2}}} \end{aligned}$$

⑧ Evaluate $\int_C (3x^2 + y^2) dx + 2xy dy$ along the circular arc C given by $x = \cos t$, $y = \sin t$ ($0 \leq t \leq \pi/2$).

Ans:-

$$\begin{aligned} \int_C (3x^2 + y^2) dx + 2xy dy & \left| \begin{array}{l} x = \cos t \\ \Rightarrow dx = -\sin t dt \\ y = \sin t \\ \Rightarrow dy = \cos t dt \end{array} \right. \\ = \int_0^{\pi/2} (3 \cos^2 t + \sin^2 t) (-\sin t) dt \\ \quad + (2 \cos t \sin t) \cos t dt \\ = \int_0^{\pi/2} (-3 \cos^2 t \sin t - \sin^3 t + 2 \cos^2 t \sin t) dt \\ = \int_0^{\pi/2} (-\cos^2 t \sin t - \sin^3 t) dt \end{aligned}$$

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$$= \int_0^{\pi/2} -\sin t (\cos^2 t + \sin^2 t) dt$$

$$= \int_0^{\pi/2} -\sin t dt = \left[\cos t \right]_0^{\pi/2}$$

$$= \cos \pi/2 - \cos 0 = \underline{\underline{-1}}$$

⑨ Find the work done in moving a particle in the force field given by $\vec{F} = (x+y)\hat{i} - x^2\hat{j} + (y+z)\hat{k}$ along the curve defined by $x^2=4y$, $z=x$, $0 \leq x \leq 2$.

Ans:-

$$W = \int_C \vec{F} \cdot d\vec{r}$$

$$= \int_C [(x+y)\hat{i} - x^2\hat{j} + (y+z)\hat{k}] \cdot [dx\hat{i} + dy\hat{j} + dz\hat{k}]$$

$$= \int_C (x+y)dx - x^2dy + (y+z)dz$$

$$= \int_0^2 \left(x + \frac{x^2}{4} \right) dx - \frac{x^3}{2} + \left(\frac{x^2}{4} + x \right) dx$$

$$= \left[\frac{x^2}{2} + \frac{x^3}{12} + \frac{x^3}{8} + \frac{x^3}{12} + \frac{x^2}{2} \right]_0^2$$

$$= \left[x^2 + \frac{x^3}{6} - \frac{x^4}{8} \right]_0^2 = 4 + \frac{8}{6} - \frac{16}{8} = 2 + \frac{8}{6} = \frac{10}{3} //$$

Given
 $x^2=4y$
 $\Rightarrow y = \frac{x^2}{4}$
 $\Rightarrow dy = \frac{x dx}{2}$
 $z=x$
 $\Rightarrow dz = dx$

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Fundamental Theorem of Line Integrals

Thm:- Suppose that $\vec{F}(x, y) = f(x, y)\hat{i} + g(x, y)\hat{j}$ is a conservative vector field in some open region D containing the points (x_0, y_0) and (x_1, y_1) and that $f(x, y)$ and $g(x, y)$ are continuous in this region.

If $\vec{F}(x, y) = \nabla\phi(x, y)$ and if C is any piecewise smooth curve that starts at (x_0, y_0) ends at (x_1, y_1) and lies in the region D , then,

$$\int_C \vec{F}(x, y) \cdot d\vec{r} = \phi(x_1, y_1) - \phi(x_0, y_0)$$

Equivalently

$$\int_C \nabla\phi \cdot d\vec{r} = \phi(x_1, y_1) - \phi(x_0, y_0)$$

OR

$$\int_a^b \vec{F} \cdot d\vec{r} = \phi(b) - \phi(a).$$

where

$$\begin{cases} a = (x_0, y_0) \text{ and} \\ b = (x_1, y_1) \end{cases}$$

RESULT :- The value of a line integral of a conservative vector field along a piecewise smooth path is independent of the path.

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Similarly, if $\vec{F}(x, y, z)$ is a vector field in 3-space then

$$\int_C \vec{F}(x, y, z) \cdot d\vec{r} = \phi(x_1, y_1, z_1) - \phi(x_0, y_0, z_0)$$

Conservation of Energy

If $\vec{F}(x, y, z)$ is a conservative force field with a potential function $\phi(x, y, z)$ then $V(x, y, z) = -\phi(x, y, z)$ is called the potential energy of the field at the point (x, y, z) .

$$\text{i.e., } W = \int_C \vec{F} \cdot d\vec{r} = \phi(x_1, y_1, z_1) - \phi(x_0, y_0, z_0) \\ = -[V(x_1, y_1, z_1) - V(x_0, y_0, z_0)]$$

Note

The work done by the field is the negative of the change in potential energy.

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① Show that the line integral $\int_{(0,2)}^{(9,0)} 4y dx + 4x dy$ is independent of the path and hence find the value of the integral.

Ans:-

We know that the line integral is independent of the path if the corresponding vector field is conservative i.e. $\nabla \times \vec{F} = 0$.

$$\text{Here } \vec{F} = 4y \hat{i} + 4x \hat{j}$$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 4y & 4x & 0 \end{vmatrix}$$

$$= \hat{i} \left[\frac{\partial}{\partial y}(0) - \frac{\partial}{\partial z}(4x) \right] - \hat{j} \left[\frac{\partial}{\partial x}(0) - \frac{\partial}{\partial z}(4y) \right] + \hat{k} \left[\frac{\partial}{\partial x}(4x) - \frac{\partial}{\partial y}(4y) \right]$$

$$= \hat{i}[0-0] - \hat{j}[0-0] + \hat{k}[4-4]$$

$$= \underline{\underline{0}} \Rightarrow \underline{\underline{\vec{F} \text{ is conservative}}}$$

Hence the integral is independent of the path.

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Now by the Fundamental Theorem

$$\int_{(1,2)}^{(4,0)} 4y dx + 4x dy = \phi(4,0) - \phi(1,2)$$

To find ϕ

$$\nabla \times \vec{F} = 0 \Rightarrow \vec{F} = \nabla \phi.$$

$$\Rightarrow 4y \hat{i} + 4x \hat{j} = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y}$$

$$\Rightarrow \frac{\partial \phi}{\partial x} = 4y, \quad \frac{\partial \phi}{\partial y} = 4x.$$

$\hookrightarrow \textcircled{1} \qquad \qquad \qquad \hookrightarrow \textcircled{2}$

Integrating $\textcircled{1}$ w.r.t 'x',

$$\phi = 4xy + f_1(y)$$

Integrating $\textcircled{2}$ w.r.t 'y',

$$\phi = 4xy + f_2(x)$$

Hence $\phi = 4xy + k.$

$$\Rightarrow \phi(4,0) = 0 + k = k$$

$$\phi(1,2) = 4 \times 1 \times 2 + k = 8 + k.$$

$$\therefore \int_{(1,2)}^{(4,0)} 4y dx + 4x dy = \phi(4,0) - \phi(1,2) \\ = k - (8 + k) = \underline{\underline{-8}}$$

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Q. Show that the line integral $\int_C (y \sin x \, dx - \cos x \, dy)$ is independent of the path.

Ans:- Given line integral is independent of the path if the corresponding vector $\vec{F}$ is conservative (irrotational or $\nabla \times \vec{F} = 0$)

$$\text{Now } \vec{F} = y \sin x \, \hat{i} - \cos x \, \hat{j}$$

$$\begin{aligned} \nabla \times \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y \sin x & -\cos x & 0 \end{vmatrix} \\ &= \hat{i} \left[\frac{\partial}{\partial y}(0) - \frac{\partial}{\partial z}(-\cos x) \right] - \hat{j} \left[\frac{\partial}{\partial x}(0) - \frac{\partial}{\partial z}(y \sin x) \right] + \hat{k} \left[\frac{\partial}{\partial x}(-\cos x) - \frac{\partial}{\partial y}(y \sin x) \right] \\ &= 0 - 0 + \hat{k} [\sin x - \sin x] \\ &= \underline{\underline{0}} \end{aligned}$$

So $\vec{F}$ is conservative.

Hence the integral is independent of the path.

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③ Show that the integral $\int_{(0,0)}^{(1, \pi/2)} (e^x \sin y \, dx + e^x \cos y \, dy)$ is independent of the path and hence evaluate the value of the integral.

Ans:- Given line integral is independent of path if the corresponding vector $\vec{F}$ is conservative.. (ie $\nabla \times \vec{F} = 0$)

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^x \sin y & e^x \cos y & 0 \end{vmatrix}$$

$$= \hat{i} \left[\frac{\partial}{\partial y}(0) - \frac{\partial}{\partial z}(e^x \cos y) \right] -$$

$$\hat{j} \left[\frac{\partial}{\partial x}(0) - \frac{\partial}{\partial z}(e^x \sin y) \right] +$$

$$\hat{k} \left[\frac{\partial}{\partial x}(e^x \cos y) - \frac{\partial}{\partial y}(e^x \sin y) \right]$$

$$= \hat{i} [0] - \hat{j} [0] + \hat{k} [e^x \cos y - e^x \cos y]$$

$$= \underline{\underline{0}}$$

$\therefore \vec{F}$ is conservative.

Hence the line integral is independent of the path.

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To evaluate line integral

$$\int_{(0,0)}^{(1,\pi/2)} e^x \sin y \, dx + e^x \cos y \, dy = \phi(1, \pi/2) - \phi(0,0) \rightarrow \text{---}$$

To find ϕ

Since $\nabla \times \vec{F} = 0 \Rightarrow \vec{F} = \nabla \phi$

$$\Rightarrow e^x \sin y \, \hat{i} + e^x \cos y \, \hat{j} = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y}$$

$$\Rightarrow \frac{\partial \phi}{\partial x} = e^x \sin y \rightarrow \text{---}$$

$$\frac{\partial \phi}{\partial y} = e^x \cos y \rightarrow \text{---}$$

Int. ① w.r.t x , $\phi = e^x \sin y + f_1(y)$

Int. ② w.r.t y , $\phi = e^x \sin y + f_2(x)$

$$\Rightarrow \phi = e^x \sin y + k$$

$$\Rightarrow \phi(1, \pi/2) = e + k$$

$$\phi(0,0) = 0 + k$$

$$\therefore \text{---} \Rightarrow \int_{(0,0)}^{(1,\pi/2)} e^x \sin y \, dx + e^x \cos y \, dy = \phi(1, \pi/2) - \phi(0,0)$$

$$= (e + k) - (k) = \underline{\underline{e}}$$